GM Crop Regulation Challenges

GS III

- Genetically modified crops contain genetic changes intended to introduce useful characteristics, such as protection against particular pests. **Bt cotton** produces proteins that target certain insects, but different Bt proteins act against different pests and do not guarantee protection against every pest species.

- When farmers repeatedly use crops containing the same pest-control proteins, insects that survive exposure may reproduce and become more common. Over time, this can reduce the crop's effectiveness, making **resistance monitoring** and appropriate pest-management practices essential.

- **Unauthorised GM seeds** create additional risks because their characteristics, performance and environmental effects may not have undergone the required regulatory assessment. Farmers may also struggle to verify seed quality or obtain reliable information when seeds circulate through informal markets.

- Agricultural biotechnology requires a **regulatory system** that can evaluate new traits, monitor cultivation and respond to emerging risks. Blocking formal approval without addressing farmers' demand for improved seeds may encourage unregulated alternatives and weaken effective oversight.

GM crops can support agriculture, but their benefits depend on scientific evaluation, pest-resistance management, reliable seed markets and regulation that responds to farmers' needs.